NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-122/3C
18 August 1989
SUPERSEDING
MIL-R-122/3B
10 June 1985

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, FILM, PRECISION,
ESTABLISHED RELIABILITY, STYLE RFP03

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for established reliability, precision, film, fixed resistors. Resistors are available within the full range of reactance, tolerance, and failure rates.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-R-122 - Resistors, Fixed, Precision, Established Reliability, General Specification For.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA (19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

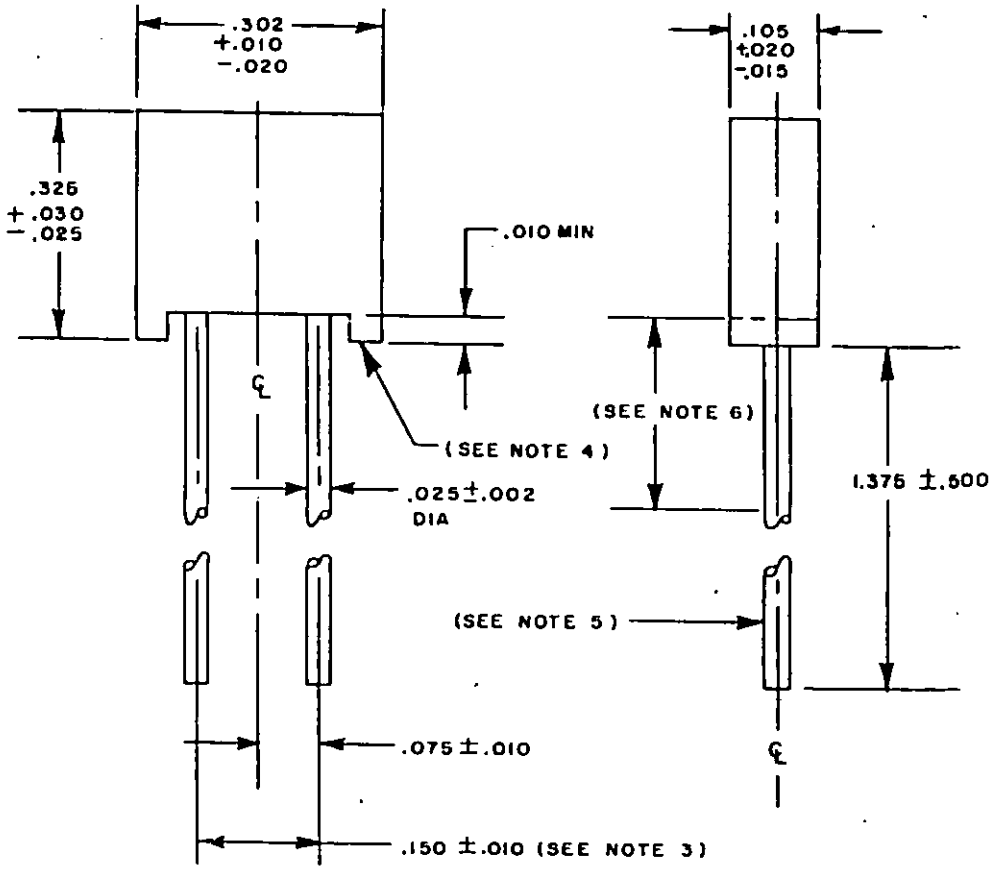
3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-122, and as specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDM 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Inches	mm	Inches	mm
.002	0.05	.105	2.67
.010	0.25	.125	3.18
.015	0.38	.150	3.81
.020	0.51	.302	7.67
.025	0.64	.325	8.26
.030	0.76	.500	12.70
.0625	1.588	1.375	34.93
.075	1.91		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The lead measurements shall be made at the point of emergence from the body.
4. Style and placement of the standoffs are optional.
5. Centerline of terminal shall coincide with the centerline of the body within ±.010 inch.
6. Resistance measuring point shall be .5 ±.125 inch for resistance values of 10Ω or more and .0625 ±.025 inch for resistance values less than 10Ω.

FIGURE 1. Fixed resistors, precision.

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3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Temperature characteristic. Resistors are available in characteristics D, E, and F.

3.3 Power rating. The power rating shall be .3 watt at 125°C (see 6.2 and 6.3). The applicable power rating curve is curve B.

3.4 Voltage rating. The continuous working voltage shall not exceed 300 volts.

3.5 Resistance. Minimum and maximum resistance values shall be as follows:

* a. Minimum resistance - 4.99 ohms.
b. Maximum resistance - 0.2 megohm.

3.6 Weight. The maximum weight shall not exceed .00221 pound (1 gram).

3.7 Overload. Apply 6.25 times rated power to the resistor for 5 seconds. When tested as specified, the applied voltage shall not exceed 450 volts.

3.8 Dielectric withstanding voltage. The applicable test voltages are as follows:

a. Atmospheric pressure - 300 volts.
b. Barometric pressure (reduced) - 200 volts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-R-122.

5. PACKAGING

5.1 Packaging. Packaging requirements shall be in accordance with MIL-R-122.

6. NOTES

6.1 Notes. The notes specified in MIL-R-122 are applicable to this specification.

6.2 Power rating at 85°C. The power rating at 85°C for 2,000 hours of life test has been established at .6 watt, 300 volts, maximum voltage. It should be noted that the failure rate level is established at the single condition of 125°C, .3 watt and 10,000 hour life test duration.

6.3 Power rating at 25°C. The power rating at 25°C for 2,000 hours of life test has been established at .75 watt, 300 volts, maximum voltage. It should be noted that the failure rate level is established at the single condition of 125°C, .3 watt and 10,000 hour life test duration.

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6.4 Changes from previous issue. The margin of this specification is marked with an asterisk to indicate where a change from the previous issue was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5905-1160)

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